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Assignment 5 — One-Dimensional Arrays
Part 1: Multiple Choice (8 Questions · 8 points)

1. For the declaration int scores[8];, which indices are valid?

☐ A) 1 through 8 ☐ B) 0 through 8
X C) 0 through 7 ☐ D) 0 through 9

2. An int array starts at base address 1000 and each int takes 4 bytes. At what address is arr[5]?

☐ A) 1005 ☐ B) 1016 **X C) 1020** ☐ D) 1024

3. Given: int a[5] = {2, 4}; What are the values of a[1] and a[3]?

X A) 4 and 0 ☐ B) 4 and garbage (unknown)
☐ C) 2 and 4 ☐ D) It does not compile

4. Which declaration is valid standard C++ (not a non-standard variable-length array)?

☐ A) int n; cin >> n; int arr[n]; ☐ B) int size = 10; int arr[size];
X C) const int N = 10; int arr[N]; ☐ D) int arr[];

5. What happens if your code writes to arr[10] when arr was declared as int arr[10];?

☐ A) A guaranteed compiler error ☐ B) The array grows by one element
☐ C) A guaranteed clear crash **X D) Compiles, but undefined behavior**

6. Given: int n[5] = {4, 9, 15, 22, 7}; int c = 0; for (int i = 0; i < 5; i++) { if (n[i] > 8) c++; } What is c?

☐ A) 2 **X B) 3** ☐ C) 4 ☐ D) 5

7. Which for loop header correctly visits an array of size SIZE from the last element to the first?

☐ A) for (int i = SIZE; i > 0; i--) **X B) for (int i = SIZE - 1; i >= 0; i--)**
☐ C) for (int i = SIZE - 1; i > 0; i--) ☐ D) for (int i = 0; i < SIZE; i--)

8. Which statement about passing an array to a function is TRUE?

☐ A) A copy is made; original is safe **X B) Can change original; const blocks it**
☐ C) The size is passed automatically ☐ D) Call it as printArray(scores[], 5)